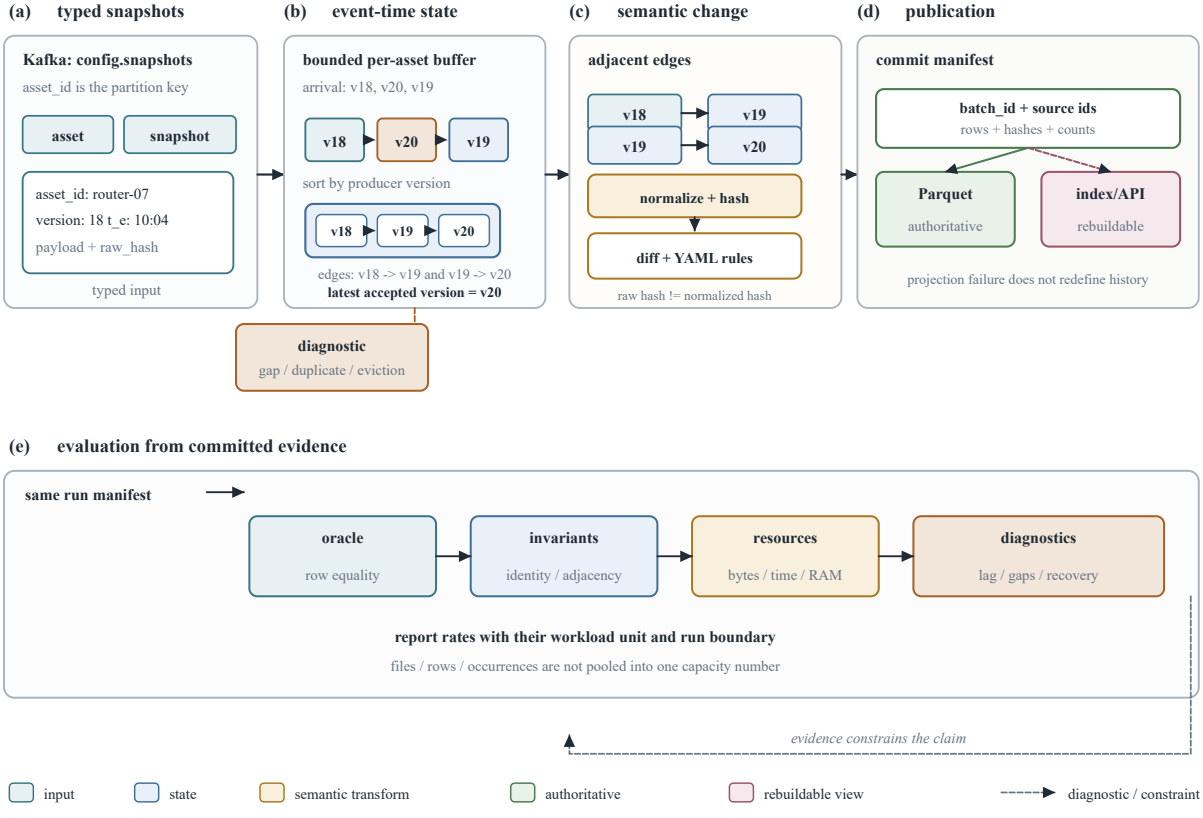

(a) typed snapshots
Kafka: config.snapshots
asset_id is the partition key
asset
snapshot
asset_id: router-07
version: 18 t_e: 10:04
payload + raw_hash
typed input
(b) event-time state
bounded per-asset buffer
arrival: v18, v20, v19
v18
v20
v19
sort by producer version
v18
v19
v20
edges: v18 -> v19 and v19 -> v20
latest accepted version = v20
diagnostic
gap / duplicate / eviction
(c) semantic change
adjacent edges
v18
v19
v19
v20
normalize + hash
diff + YAML rules
raw hash != normalized hash
(d) publication
commit manifest
batch_id + source ids
rows + hashes + counts
Parquet
authoritative
index/API
rebuildable
projection failure does not redefine history
(e) evaluation from committed evidence
same run manifest
oracle
row equality
invariants
identity / adjacency
resources
bytes / time / RAM
diagnostics
lag / gaps / recovery
report rates with their workload unit and run boundary
files / rows / occurrences are not pooled into one capacity number
evidence constrains the claim
input
state
semantic transform
authoritative
rebuildable view
diagnostic / constraint